

5. Novon na tuljivo v mag. polju

dve tuljivi bono postavili eno v drugo. Glavnejšo bono pustali tož  
v manjšo & bo inducirala tož in tobo tudi mag. novon  
v manjšo tuljivo bono tudi opustili enomeni tož, tobo & bo  
ročela uteti novon:

$$\vec{M} = \vec{p} \times \vec{B}$$

↑  
N'I'S'

novon bono izvenili & uterjajo in iz:

$$N'I'S' \frac{\mu_0}{l} \frac{NI}{2} = mgr$$

izvenili  $\mu_0$

nevarnosti: velik enomeni tož do 10A! - najbolj nevarno  
tuljivo & bo tudi segreva in bo luha močja tudi po času  
- najbolj pogosto  
povi odprte kontakte!

\* mentor na pi 6.

$$\mu_0 = \frac{mgr \cdot l}{NI N'I'S'}$$

razlika last.

$$m = 0,0005 \text{ kg}$$

$$\pm 0,00002$$

$$\mu_1 = 1,235 \cdot 10^{-6} [\text{nebe}] \quad \checkmark$$

$$\text{kg m}^2 \text{ s}^{-2} \text{ A}^{-2}$$

$$r = 0,04 \pm 0,001 \text{ m}$$

$$B = \mu_0 \frac{NI}{l}$$

$$\frac{\text{kg m}^2}{\text{s}^2 \text{ A}^2}$$

$$l = 0,562 \text{ m} \pm 0,02$$

$$[T] = \left[ \frac{\text{A}}{\text{m}} \right]$$

↑  
 $\frac{T \text{ m}}{A}$

$$T = \frac{\text{kg}}{\text{s}^2 \text{ A}}$$

$$N_v = 24 \cdot 4 = 96$$

$$N' = 150$$

$$S' = 0,004 \text{ m}^2 \pm 0,0002 \text{ m}^2$$

$$\frac{\text{kg m}}{\text{s}^2 \text{ A}^2}$$

CA 101b1



Imeli smo večkrat in malo tuljavo ter meritev med 1x meritev in 2x.

1x:

$$\textcircled{1} I_v = 5,2 \text{ A} \quad I_m = 0,295 \text{ A}$$

$$\textcircled{2} I_v = 6,0 \text{ A} \quad I_m = 0,265 \text{ A}$$

2x:

$$\textcircled{3} I_v = 3,6 \text{ A} \quad I_m = 0,215 \text{ A}$$

$$\textcircled{4} I_v = 3,0 \text{ A} \quad I_m = 0,26 \text{ A}$$

$$(1-10 \text{ A}) \quad (0-0,3 \text{ A})$$

napake:

$$\delta I_v = 0,2 \text{ A}$$

$$\delta I_m = 0,006 \text{ A}$$

$$m = 0,5 \text{ g} \pm 0,02 \text{ g} \quad 4\%$$

$$r = 0,04 \pm 0,001 \text{ m} \quad \text{rel. } 3\%$$

$$l = 0,562 \text{ m} \pm 0,02 \text{ m} \quad 1\%$$

$$N_v = 24 \cdot 4 = 96$$

$$N' = 150$$

$$S' = 0,004 \text{ m}^2 \pm 0,0002 \text{ m}^2 \quad 5\%$$

$$N' I' S' \mu_0 \frac{N I}{l} = m g r$$

$$\Rightarrow \mu_0 = \frac{m g r \cdot l}{N I N' I' S'} =$$

Poimenovanja

$$\begin{cases} I_v = I \\ I_m = I' \end{cases}$$

$$\textcircled{1} \mu_0 = \frac{5 \cdot 10^{-4} \text{ kg} \cdot 9,81 \frac{\text{m}}{\text{s}^2} \cdot 0,04 \text{ m} \cdot 0,562 \text{ m}}{96 \cdot 5,2 \text{ A} \cdot 150 \cdot 0,295 \text{ A} \cdot 0,004 \text{ m}^2} = 1,235 \cdot 10^{-6} \frac{\text{Tm}}{\text{A}}$$

$$\mu_{\text{err}} = \underbrace{4\%}_m + \underbrace{0,1\%}_g + \underbrace{3\%}_r + \underbrace{1\%}_l + \underbrace{4\%}_{I_v} + \underbrace{2\%}_{I_m} + \underbrace{5\%}_{S'}$$

$$= 19\% \Rightarrow 20\%$$

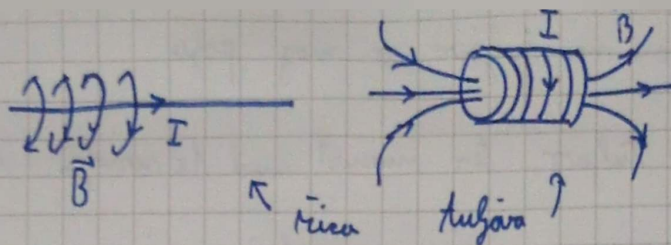
$$\bar{\mu}_{\text{err}} = \frac{\sqrt{\sum \sigma^2}}{n} = 10\%$$

$$\left[ \begin{aligned} \mu_1 &= 1,24 \cdot 10^{-6} \frac{\text{Tm}}{\text{A}} (1 \pm 0,2) = (1,2 \pm 0,2) \cdot 10^{-6} \frac{\text{Tm}}{\text{A}} \\ \mu_2 &= 1,20 \cdot 10^{-6} (1 \pm 0,2) \frac{\text{Tm}}{\text{A}} = (1,2 \pm 0,2) \cdot 10^{-6} \frac{\text{Tm}}{\text{A}} \\ \mu_3 &= 1,25 \cdot 10^{-6} (1 \pm 0,2) \frac{\text{Tm}}{\text{A}} = (1,3 \pm 0,2) \cdot 10^{-6} \frac{\text{Tm}}{\text{A}} \\ \mu_4 &= 1,23 \cdot 10^{-6} (1 \pm 0,2) \frac{\text{Tm}}{\text{A}} = (1,2 \pm 0,2) \cdot 10^{-6} \frac{\text{Tm}}{\text{A}} \end{aligned} \right]$$

$$\bar{\mu} = 1,225 \cdot 10^{-6} \frac{\text{Tm}}{\text{A}} (1 \pm 0,1) = (1,2 \pm 0,1) \cdot 10^{-6} \frac{\text{Tm}}{\text{A}}$$

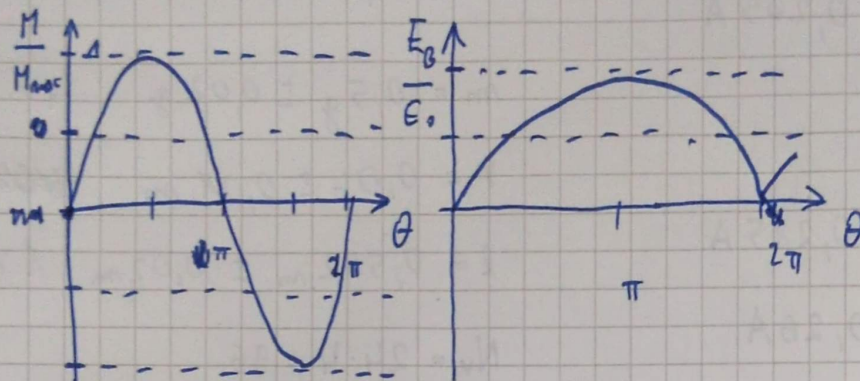


2.1:

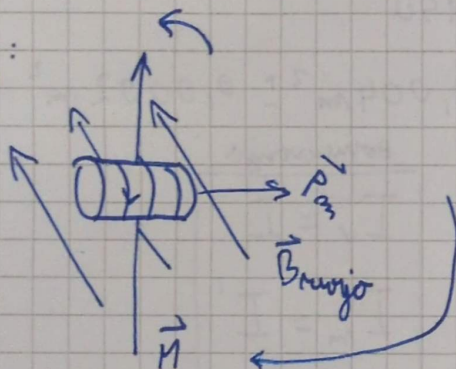


2.2:  $B = \frac{\mu_0 I}{2\pi r}$  ← rea neskonci vodiak  $\frac{1}{r} \rightarrow$  linearno obrato sorvornu

2.3:



2.4:



$\vec{M} = \vec{p}_m \times \vec{B}$  ← deluje pravokotno

2.5: edspust bi obe tufjari monoli postaviti tako da je  $\vec{p}_m$  v isto smer kot kemljan  $\vec{B}$ . V vozi so mag. polja bila v redni  $10^{-3} \text{ T}$  kemljan pa samo 1% napake doba kemljan polje v eno smer  $10^{-5} \text{ T}$

2.6:  $\frac{2r}{l} \approx (0,16 \pm 0,02)$